

QSS 20 Final Project Milestone 1

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1 Project Option

Describe the final project you are pursuing.

- Senior Thesis / Own Data: I am pursuing a project using my own data on the mismatch between where the burden of neurological and neurodegenerative disease falls and where the workforce capable of diagnosing and managing that disease exists. The unit of analysis is the country; the end goal is for the project to combine three sources: the WHO/World Federation of Neurology *Atlas: Country Resources for Neurological Disorders* (2nd ed., 2017), which reports neurological workforce density and service availability for 133 countries and territories; the IHME Global Burden of Disease (GBD) study, which supplies country-year estimates of neurological disease burden in DALYs, prevalence, and incidence; and the Demographic and Health Surveys (DHS), which supply population-level demographic and health-system covariates and, through the geographic datasets, cluster-level information on distance to care. This milestone works with the Atlas data, which I have already extracted and cleaned; the GBD and DHS extracts are in progress, and I am unsure to the extent I will be able to use them. The project is Part 1 of a two-part design: the country-level access gradient estimated in QSS 20 is intended to enter a separate household-level analysis of informal caregiving reliance as a contextual predictor for QSS 45.

2 Your project's main questions

Write 1 paragraph on the main questions you hope to answer.

I have three main questions that are all connected that I hope to answer. First, how large is the gap between neurological disease burden and neurological workforce capacity, and how does that gap vary across World Bank income groups and WHO regions? Second, is the gap better characterized as a national scarcity problem or as a distributional problem within countries; do countries with low aggregate workforce density also concentrate what workforce they have in capital cities, so that the effective access gap for a rural resident is substantially worse than the national density figure implies? Third, once income group and demographic structure are accounted for, does residual variation in workforce density predict anything about how neurological need is met outside the formal health system? I

do not think that it will be possible to fully identify a causal effect of workforce density on outcomes; the goal is to construct a defensible, reproducible measure of the burden–capacity gap and to characterize its structure, which is a prerequisite for the household-level question, but it is not its answer.

3 Your project’s data sources/relevant fields

Give a 1 paragraph tentative summary of potentially relevant data and fields; you do not need to know the data exhaustively yet; this can just be based on codebook or reading in some sample data/viewing its head.

The WHO/WFN Neurology Atlas is the main workforce source, and I extracted two tables directly from the PDF. The first is the participating-country annex, which yields 133 rows with fields `country`, `who_region`, and `wb_income_group`; this serves as the main aspect onto which the other sources merge. The second is the results tables (Tables 2–3, Figures 11–12, 15–16), which give, by income group and by WHO region, the median total neurological workforce per 100 000 population, disaggregated medians for adult neurologists, neurosurgeons, and child neurologists, the number of responding countries behind each median, and the share of countries reporting that neurologists practice in the capital city, in other urban areas, and in rural areas. The Atlas is a survey of ministry-of-health focal points, so response counts vary by item (114 countries for workforce density, 93 for child neurologists), and this variation has to be carried through the analysis rather than ignored. The IHME GBD extract will supply `location_name`, `year`, `cause_name`, `measure_name` (DALYs, prevalence, incidence), `metric_name` (rate per 100 000 and number), and `val/upper/lower`, restricted to the Level 2 neurological disorders cause group and its Level 3 children (stroke, dementias, epilepsy, Parkinson disease, migraine) for all countries. The GBD file I currently have is a global all-cause summary and is not usable at the country level; it is being re-pulled at country resolution. DHS will supply population age structure, urban/rural residence, and health-service utilization indicators, plus GPS cluster coordinates for the distance-to-facility component; the DHS application is pending approval.

4 Create one starter visualization

Create two visualizations using the skills you’ve gained from the Pandas and matplotlib modules. Embed them here.

Both figures below are built from the Atlas data described above using pandas and matplotlib. Figure 1 establishes the aggregate scarcity gradient. The median total neurological workforce is 0.1 per 100 000 population in low-income countries against 7.1 in high-income countries, which is a 71-fold difference, and the gap is wider still for adult neurologists specifically (0.03 versus 4.75, roughly 158-fold). Two of the four income groups sit below the global median of 3.1, and the low-income median is close enough to zero that a country-level regression on this outcome will be badly skewed and likely need a log or rank transformation.

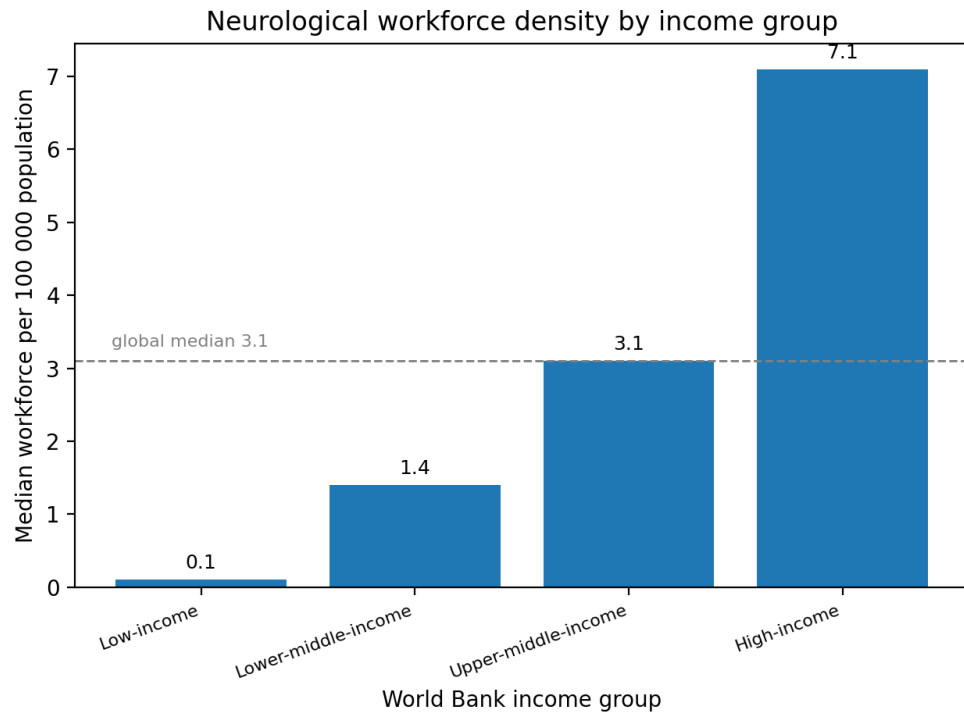


Figure 1: Median neurological workforce per 100 000 population by World Bank income group. Source: WHO/WFN Neurology Atlas, 2nd ed. (2017), Fig. 12; $N = 114$ responding countries.

Figure 2 helps to answer the second question for this project. Reading across income groups, the share of countries reporting neurologists in the capital city is almost flat, 95% to 100% everywhere, while the share reporting any neurologist practicing in rural areas falls from 45% in high-income countries to 0% in low-income countries. In other words, the national density figures in Figure 1 understate the problem: in the poorest countries the entire specialist workforce is urban, so a rural resident faces an effective density of zero regardless of what the national median says. This is the specific pattern the DHS geographic data is intended to quantify if I were to gain access.

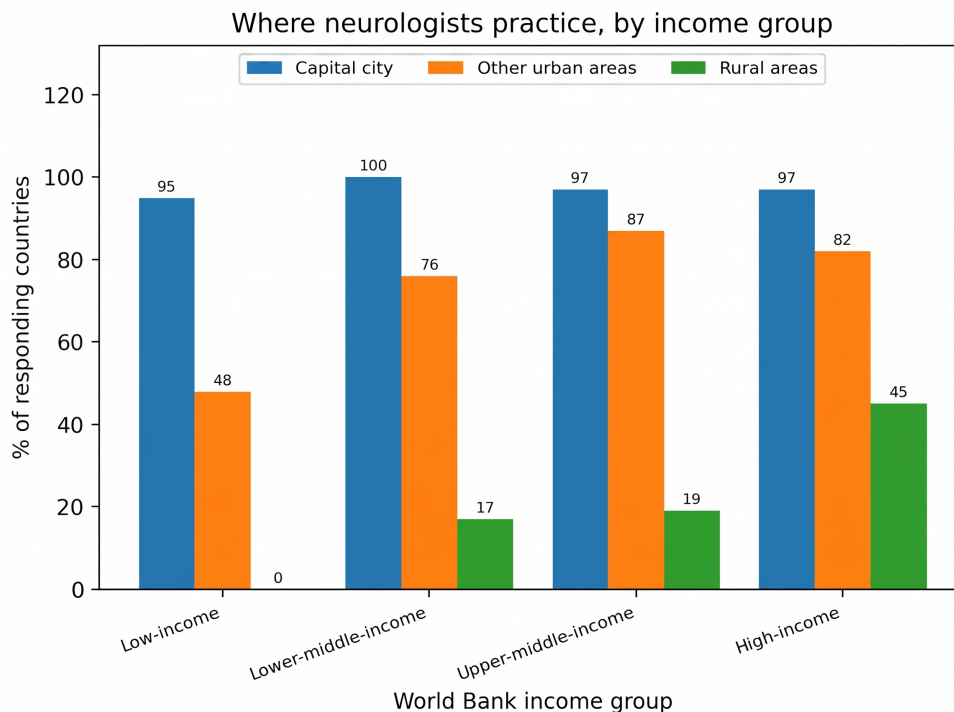


Figure 2: Share of responding countries reporting that neurologists practice in each setting, by World Bank income group. Source: WHO/WFN Neurology Atlas, 2nd ed. (2017), Fig. 16; $N = 114$ responding countries.

5 Learn from an earlier project on your research topic

Read the report for [the provided example on your project option from Fall 2022](#) (for felony sentencing, read one of the two). Describe one lesson you take away from their report and one weakness of their approach upon which you will try to improve.

I read the descriptive comparison example on felony sentencing: Fink, Jin, Punaini and Tolat's study of racial disparities in Cook County bail decisions. The topic is unrelated to mine, but the structure is nearly identical: a descriptive comparison of outcomes across

groups, drawn from administrative data, where the quantity of interest is a gap and the central threat is confounding the data cannot resolve.

The lesson I take away is about their choice of comparison units: they never compare Black and White defendants in the aggregate. Every analysis first subsets to a specific offense category, Armed Robbery, Homicide, Unlawful Use of Weapon, Robbery, Battery, on the reasoning that a pooled comparison would mostly reflect differences in charge composition rather than differential treatment of similar cases. Equally useful is their insistence on reporting a disparity on two scales at once. In their Table 8, bond-increase rates differ across race by as little as 0.0006 in absolute terms while differing by factors of 1.25 to 8 in relative terms; on either scale alone the finding would be badly misleading; I intend to import both habits. Workforce density should never be compared across countries without conditioning on income group, and the burden–capacity gap should be reported in absolute and relative terms together, since the low versus high income contrast in adult neurologist density is a 4.7-point absolute gap per 100 000 population and simultaneously a 158-fold relative one.

The weakness I want to improve on is that they identify their central confounder and then proceed as though naming it discharged the obligation. Criminal history is unavailable in their data, and they say so plainly, in the Methods section, and again in the Limitations section, but it never re-enters the analysis. There is no bounding exercise, no sensitivity analysis, and no attempt to ask how large the unmeasured confounding would have to be to erase the disparities they report. The reader is left holding a stated caveat and a table of point estimates with no way to reconcile them. A related gap: they report p -values for Table 2 but not for Tables 3 through 8, and the relative differences in Table 8 rest on rates between 0.0001 and 0.0015, so an eight-fold ratio may be built on a handful of cases with no counts or intervals shown. Their inner join also retains only 45% of case participants, a selection they speculate about but never test.

My analysis faces the structurally identical problem: national income drives both neurological workforce capacity and measured disease burden, and measured burden additionally depends on the diagnostic capacity that is my exposure. I plan to improve on their treatment in two concrete ways. First, every estimate will carry its denominator: Atlas response counts already vary from 93 to 114 countries by item, and my merged file will keep a per-country coverage indicator for each source rather than silently dropping incomplete rows. Second, income will be modeled and bounded rather than merely disclosed: stratified estimates within income group, plus an explicit statement of how large a confounding effect would need to be to overturn the observed gap. Naming a confounder in the limitations section is not the same as accounting for it.

6 Your project’s anticipated challenges

Conduct a 1 paragraph pre-mortem on challenges your project may face – e.g., measuring something, dealing with missing data, disentangling causality.

There are a couple of challenges that are present. The first is measurement: “neurological workforce” in the Atlas is a self-reported count from a ministry focal point, with no auditing, no full-time-equivalent adjustment, and no guarantee that countries applied the

glossary definitions consistently, so cross-country comparisons are noisier than the medians suggest. The second is missingness, since it is likely that it is not random: the countries that failed to respond to the Atlas survey are likely those with the weakest health information systems and therefore the weakest neurological services, which would bias any complete-case analysis toward understating the gap. I plan to handle this with explicit missingness indicators and sensitivity analysis rather than imputation that assumes MAR. The third is temporal and unit alignment: the Atlas reflects 2014–2015 data collection, GBD estimates are annual through 2023, and DHS surveys are fielded country-by-country on irregular cycles, so any merged panel will require decisions about which year to use the most and will have broken coverage. Country-name harmonization across three sources with three different naming conventions is a related issue. The fourth is causality since workforce density, disease burden, and national income are interconnected, income affects both capacity and measured burden, and measured burden depends on diagnostic capacity, which is the exposure, that any estimate framed as the effect of workforce on outcomes would be uninterpretable. The project is as a result uses the burden–capacity gap as a construct to be measured well rather than a treatment to be estimated.